

A Survey of Algorithmic Recourse: Contrastive Explanations and Consequential Recommendations

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Machine learning is increasingly used to inform decision making in sensitive situations where decisions have consequential effects on individuals' lives. In these settings, in addition to requiring models to be accurate and robust, socially relevant values such as fairness, privacy, accountability, and explainability play an important role in the adoption and impact of said technologies. In this work, we focus on *algorithmic recourse*, which is concerned with providing explanations and recommendations to individuals who are unfavorably treated by automated decision-making systems. We first perform an extensive literature review, and align the efforts of many authors by presenting unified definitions, formulations, and solutions to recourse. Then, we provide an overview of the prospective research directions toward which the community may engage, challenging existing assumptions and making explicit connections to other ethical challenges such as security, privacy, and fairness.

CCS Concepts: • **Networks** → *Network properties; Network algorithms;*

Additional Key Words and Phrases: Algorithmic recourse, contrastive explanations and consequential recommendations

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1 INTRODUCTION

Consider the following setting: a 28-year-old female professional working as a software engineer seeks a mortgage to purchase a home. Consider further that the loan-granting institution (e.g., bank) uses a binary classifier and denies the individual the loan based on her attributes. Naturally, in this context, answering the following questions become relevant to the individual:

Q1: Why was I rejected the loan?

Q2: What can I do to get the loan in the future?

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In the setting of the preceding example, unless the policy of the bank is relaxed, the individual must expend effort to change their situation to be favorably treated by the decision-making system. Examples such as this are prevalent not only in finance [20, 168] but also in justice (e.g., pretrial bail) [10], healthcare (e.g., prescription of life-altering medication) [24, 28, 91], and other settings (e.g., hiring) [50, 149, 169, 207] broadly classified as *consequential decision-making* settings [20, 35, 52, 113]. Given the rapid adoption of automated decision-making systems in these settings, designing models that not only have high objective accuracy but also afford the individual with *explanations* and *recommendations* to favorably change their situation is of paramount importance, and even argued to be a legal necessity (GDPR [231]). This is the concern of *algorithmic recourse*.

Contributions. Our review brings together the plethora of recent works on algorithmic recourse. In reviewing the literature, we identify opportunities to consolidate definitions, construct technical baselines for future comparison, and situate recourse in the broader ethical **machine learning (ML)** literature. The primary contribution is thus a unified overview of the definitions (Section 2), formulations (Section 3), and technical solutions (Section 4) for computing (*nearest*) *contrastive explanations* and (*minimal*) *consequential recommendations*, across a broad range of setups. A high-level summary is curated and presented in Table 1. Clearly distinguishing between the preceding two recourse offerings and presenting their often overlooked different technical treatment is a common theme throughout this work. Additionally, we identify a number of prospective research directions which challenge the assumptions of existing setups, and present extensions to better situate recourse in the broader ethical ML literature (Section 5).

Who is this article for? This article aims to engage different audiences: practitioners (P), researchers (R), and legal scholars (L). Section 2 builds on the rich literature in the philosophy of science and presents a clear distinction between contrastive explanations and consequential recommendations based on the causal history of events (R, L). Table 1 presents an overview of 50+ technical papers that offer recourse in a broad range of setups (P, R). Section 3 formulates the recourse problem as constrained optimization and presents the variety of constraints needed to model real-world settings (R). Section 4 presents the difficulty in solving for recourse in general settings and summarizes existing approximate solutions that trade off various desirable properties (P, R, L). Finally, Section 5 argues that recourse should be considered in relation to other ethical ML considerations (L) and outlines future research directions (R).

Scope. To adequately present the material, we provide brief introductions to various related topics, such as explainable ML (or XAI), causal and counterfactual inference, and optimization, among others. Our work is not meant to be a comprehensive review of these topics and merely aims to situate algorithmic recourse in the context of these related fields. For the interested reader, an in-depth discussion on the social, philosophical, cognitive, and psychological aspects of explanations can be found elsewhere [21, 36, 157, 226]. Other works of interest are available on XAI [1, 2, 32, 68, 79, 86, 94, 139, 150, 163], causal and counterfactual inference, [82, 95, 166, 185, 186, 206, 245], and optimization [33, 174, 214].

2 BACKGROUND

2.1 Recourse Definitions

In its relatively young field, *algorithmic recourse* has been defined as, for example, “an actionable set of changes a person can undertake in order to improve their outcome” [109], “the ability of a person to obtain a desired outcome from a fixed model” [223], and “the systematic process of reversing unfavorable decisions by algorithms and bureaucracies across a range of counterfactual scenarios” [226]. Similarly, *legal recourse* pertains to actions by individuals or corporations to remedy legal difficulties [235]. Parallel to this, *explanations* aim to, at least in part, assist data-subjects

to “understand what could be changed to receive a desired result in the future” [234]. This plurality of overlapping definitions, and the evolving consensus therein, motivates us to attempt to present a unified definition under the umbrella of recourse.

Intuitively, the commonality shared among these definitions is the desire to assist individuals who are negatively affected by automated decision-making systems to improve their circumstances (captured in the individual’s feature set) to overcome the adverse decision (from the predictive model). Such assistance may take on the form of *explanations* (where they need to get to) or *recommendations* (what actions should they perform to get there).

Therefore, we submit that recourse can be achieved by an affected individual if they can understand [69, 109, 234] and accordingly act [114, 115] to alleviate an unfavorable situation, thus exercising temporally extended agency [226]. Concretely, in the example of the previous section, recourse is offered when the loan applicant is given answers to both questions: provided *explanation(s)* as to why the loan was rejected (Q1) and offered *recommendation(s)* on how to obtain the loan in the future (Q2). In the following, we describe the similarities and often overlooked differences between these questions and the different set of assumptions and tools needed to sufficiently answer each in general settings. Such a distinction is made possible by looking at the context from a causal perspective, which we summarize next.

2.2 Recourse and Causality

2.2.1 Contrastive Explanations. In an extensive survey of the social science literature, Miller [157] concluded that when people ask “Why P?” questions (e.g., Q1), they are typically asking “Why P rather than Q?” where Q is often implicit in the context [101, 199]. A response to such questions is commonly referred to as a *contrastive explanation* and is appealing for two reasons. First, contrastive *questions* provide a “window” into the questioner’s mental model, identifying what they had expected (i.e., Q, the contrast case), and thus the *explanation* can be better tuned toward the individual’s uncertainty and gap in understanding [156]. Second, providing contrastive explanations may be “simpler, more feasible, and cognitively less demanding” [156] than offering recommendations.

2.2.2 Consequential Recommendations. Providing an affected individual with recommendations (e.g., response to Q2) amounts to suggesting a set of actions (a.k.a. *flipsets* [223]) that should be performed to achieve a favorable outcome in the future. In this regard, several works have used contrastive explanations to directly infer actionable recommendations [109, 209, 223, 234] where actions are considered as independent shifts to the feature values of the individual (P) that results in the contrast (Q). Recent work, however, has cautioned against this approach, citing the implicit assumption of independently manipulable features as a potential limitation that may lead to sub-optimal or infeasible actions [21, 114, 146, 226]. To overcome this, Karimi et al. [114] suggest that actions may instead be interpreted as *interventions* in a causal model of the world in which actions will take place and not as independent feature manipulations derived from contrastive explanations. Formulated in this manner, for example, using a **structural causal model (SCM)** [184], the downstream effects of interventions on other variables in the model (e.g., descendants of the intervened-upon variables) can directly be accounted for when recommending actions [21]. Thus, a recommended set of actions for recourse, in a world governed by an SCM, are referred to as *consequential recommendations* [114].

2.2.3 Clarifying Terminology: Contrastive, Consequential, and Counterfactual. To summarize, recourse explanations are commonly sought in a contrastive manner, and recourse recommendations can be considered as interventions on variables modeled using an SCM. Thus, we can rewrite the two recourse questions as follows:

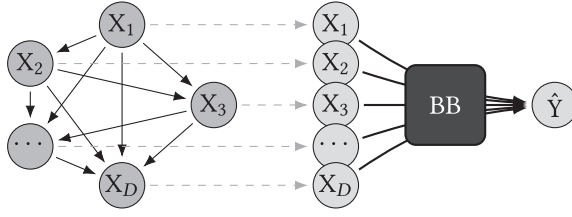


Fig. 1. Variables $\{X_i\}_{i=1}^D$ capture observable characteristics of an individual that are fed into a black-box (BB) decision-making system (*mechanism*) yielding the prediction, $\hat{Y}$. Consequential recommendations are interventions on a causal model of the world in which actions take place and may have downstream effects on other variables before being passed into the BB. Conversely, contrastive explanations are obtained via interventions on the BB inputs that can be seen as independent feature shifts that do not affect other variables. Unlike the latter, generating the former relies on accurate knowledge of the SCM or causal graph as a model of nature itself (both of which are non-identifiable from observational data alone [188]). The result of performing a consequential recommendation is a contrastive explanation (see Section 2.2 and Section 3 for descriptive and technical details, respectively).

Q1: What profile would have led to receiving the loan?

Q2: What actions would have led me to develop this profile?¹

Viewed in this manner, both contrastive explanations and consequential recommendations can be classified as a *counterfactual* [151], in that each considers the alteration of an entity in the history of the event P, where P is the undesired model output. Thus, responses to Q1 (respectively, Q2) may also be called *counterfactual explanations* (respectively, *counterfactual recommendations*), meaning what could have been (respectively, what could have been done) [36].²

To better illustrate the difference between contrastive explanations and consequential recommendations, we refer to Figure 1. According to Lewis [137, p. 217], “to explain an event is to provide some information about its causal history.” Lipton [138, p. 256] argues that to explain why P rather than Q, “we must cite a causal difference between P and not-Q, consisting of a cause of P and the absence of a corresponding event in the history of not-Q.” In the algorithmic recourse setting, because the model outputs are determined by its inputs (which temporally precede the prediction), the input features may be considered as the causes of the prediction. The determining factor in whether one is providing contrastive explanations as opposed to consequential recommendations is thus the level at which the *causal history* [203] is considered: whereas providing explanations only requires information on the relationship between the model inputs, $\{X_i\}_{i=1}^D$, and predictions, $\hat{Y}$, recommendations require information as far back as the causal relationships among inputs. The reliance on fewer assumptions [138, 157] thus explains why generating recourse explanations is easier than generating recourse recommendations [156, 234].

Next, we summarize the technical literature in Table 1 (subsequent surveys can be found elsewhere [117, 215, 227]). We find that most of the recourse literature has focused on generating contrastive explanations rather than consequential recommendations (c.f., Section 2). Differentiable models are the most widely supported class of models, and many constraints are only sparsely

¹A common assumption when offering recommendations is that the world is stationary; thus, actions that *would have* led me to develop this profile had they been performed in the past *will* result in the same were they to be performed now. This assumption is challenged in other works [194, 226] and discussed further in Section 5.3.

²Note that “some researchers tend to either collapse or intentionally distinguish contrastive from counterfactual reasoning despite their conceptual similarity” [215], adding to confusion. For cross-disciplinary reviews, please refer to other works [156, 157, 215].

Table 1. Overview of Recourse Algorithms for Consequential Decision-Making Settings

	Algorithm	Formulation													Solution							
		Model				Actionability		Plausibility			Extra		Datatypes				Tools	Access	Properties			Code
		TB	KB	DF	OT	uncond.	cond.	dom.	dens.	proto.	diver.	spar.				opt.			cov.	rtm.		
(2014.03)	SEDC [148]	●	●	●	●	○	○	○	○	○	○	○	○	○	heuristic	query	●	○	●	●		
(2015.08)	OAE [55]	●	○	○	○	●	●	●	●	○	○	○	○	○	ILP	white-box	●	○	●	○		
(2016.05)	HCLS [128, 130]	●	●	○	○	○	○	○	○	○	○	○	○	○	grad opt/heuristic	gradient/query	●	○	●	●		
(2017.06)	Feature Tweaking [219]	●	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	white-box	○	○	○	○		
(2017.11)	CF Expl. [234]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2017.12)	Growing Spheres [132]	●	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2018.02)	CEM [63]	○	○	○	○	○	○	○	○	○	○	○	○	○	FISTA	class prob.	○	○	○	○		
(2018.02)	POLARIS [249]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	gradient	○	○	○	○		
(2018.05)	LORE [93]	○	○	○	○	○	○	○	○	○	○	○	○	○	gen alg + heuristic	query	○	○	○	○		
(2018.06)	Local Foil Trees [224]	●	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2018.09)	Actionable Recourse [223]	○	○	○	○	○	○	○	○	○	○	○	○	○	ILP	white-box	●	○	○	○		
(2018.11)	Weighted CFs [90]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2019.01)	Efficient Search [205]	○	○	○	○	○	○	○	○	○	○	○	○	○	MILP	white-box	○	○	○	○		
(2019.04)	CF Visual Expl. [89]	○	○	○	○	○	○	○	○	○	○	○	○	○	greedy search	white-box	○	○	○	○		
(2019.05)	MACE [113]	○	○	○	○	○	○	○	○	○	○	○	○	○	SAT	white-box	○	○	○	○		
(2019.05)	DiCE [167]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2019.05)	CERTIFAI [209]	●	○	○	○	○	○	○	○	○	○	○	○	○	gen alg	query	○	○	○	○		
(2019.06)	MACEM [64]	○	○	○	○	○	○	○	○	○	○	○	○	○	FISTA	query	○	○	○	○		
(2019.06)	Expl. using SHAP [193]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2019.07)	Nearest Observable [241]	○	○	○	○	○	○	○	○	○	○	○	○	○	brute force	dataset	○	○	○	○		
(2019.07)	Guided Prototypes [225]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt/FISTA	gradient/query	○	○	○	○		
(2019.07)	REVISE [109]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2019.08)	CLEAR [242]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2019.08)	MC-BRP [142]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2019.09)	FACE [189]	○	○	○	○	○	○	○	○	○	○	○	○	○	graph + heuristic	query	○	○	○	○		
(2019.09)	Equalizing Recourse [96]	○	○	○	○	○	○	○	○	○	○	○	○	○	ILP/heuristic	white-box/query	○	○	○	○		
(2019.10)	Action Sequences [190]	○	○	○	○	○	○	○	○	○	○	○	○	○	program synthesis	class prob.	○	○	○	○		
(2019.10)	C-CHVAE [183]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt + heuristic	query + gradient	○	○	○	○		
(2019.11)	FOCUS [143]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt + heuristic	white-box	○	○	○	○		
(2019.12)	Model-based CFs [146]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2019.12)	LIME-C/SHAP-C [191]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2019.12)	EMAP [43]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	dataset/query	○	○	○	○		
(2019.12)	PRINCE [84]	○	○	○	○	○	○	○	○	○	○	○	○	○	graph + heuristic	query	○	○	○	○		
(2019.12)	LowProFool [18]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2020.01)	ABELE [92]	○	○	○	○	○	○	○	○	○	○	○	○	○	gen alg + heuristic	query + data	○	○	○	○		
(2020.01)	SHAP-based CFs [77]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2020.02)	CEML [11–13]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt/heuristic	gradient/query	○	○	○	○		
(2020.02)	MINT [114]	○	○	○	○	○	○	○	○	○	○	○	○	○	SAT	white-box	○	○	○	○		
(2020.03)	ViCE [87]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2020.03)	Plausible CFs [22]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt + gen alg	dataset	○	○	○	○		
(2020.04)	SEDC-T [229]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2020.04)	MOC [56]	○	○	○	○	○	○	○	○	○	○	○	○	○	gen alg	query	○	○	○	○		
(2020.04)	SCOUT [237]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2020.04)	ASP-based CFs [29]	○	○	○	○	○	○	○	○	○	○	○	○	○	answer-set prog.	query	○	○	○	○		
(2020.05)	CBR-based CFs [118]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query + data	○	○	○	○		
(2020.06)	Survival Model CFs [124]	○	○	○	○	○	○	○	○	○	○	○	○	○	gen alg	query	○	○	○	○		
(2020.06)	Probabilistic Recourse [115]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt/brute force	gradient/query	○	○	○	○		
(2020.06)	C-CHVAE [182]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2020.07)	FRACE [250]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2020.07)	DACE [110]	○	○	○	○	○	○	○	○	○	○	○	○	○	MILP	white-box	○	○	○	○		
(2020.07)	CRUDS [69]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient/data	○	○	○	○		
(2020.07)	Gradient Boosted CFs [5]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	data	○	○	○	○		
(2020.08)	Gradual Construction [111]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	class prob.	○	○	○	○		
(2020.08)	DECE [46]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt	gradient	○	○	○	○		
(2020.08)	Time Series CFs [16]	○	○	○	○	○	○	○	○	○	○	○	○	○	heuristic	query	○	○	○	○		
(2020.08)	PermuteAttack [100]	○	○	○	○	○	○	○	○	○	○	○	○	○	gen alg	query	○	○	○	○		
(2020.10)	Fair Causal Recourse [233]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt/brute force	gradient/query	○	○	○	○		
(2020.10)	Recourse Summaries [195]	○	○	○	○	○	○	○	○	○	○	○	○	○	itemset mining alg	query	○	○	○	○		
(2020.10)	Strategic Recourse [45]	○	○	○	○	○	○	○	○	○	○	○	○	○	Nelder-Mead	query	○	○	○	○		
(2020.11)	PARE [202]	○	○	○	○	○	○	○	○	○	○	○	○	○	grad opt + heuristic	query	○	○	○	○		

Ordered chronologically, we summarize the *goal*, *formulation*, *solution*, and *properties* of each algorithm. Symbols are used to indicate supported settings in the experimental section of the article (●), settings that are natural extensions of the presented algorithm (◐), settings that are partially supported (◑), and settings that are not supported (○). The models cover a broad range of tree-based (TB), kernel-based (KB), differentiable (DF), or other (OT) types. Actionability constraints (unconditional or conditional), plausibility constraints (domain-, density-, and prototypical-consistency), and additional constraints (diversity, sparsity) are also explored. While the primary datatypes used in consequential settings are tabular  (involving a mix of numeric, binary, categorical, and ordinal variables), we also include additional works that generate recourse for non-tabular (images  and document ) datasets. Furthermore, papers that present analysis of such properties as optimality (opt.), coverage (cov.), and runtime complexity (rtm.) are specified in the table. Finally, we make note of those papers that provide *open source* implementations of their algorithm.

supported (c.f., Section 3). All tools generate solutions that to some extent trade off desirable requirements, such as optimality, perfect coverage, efficient runtime, and access (c.f., Section 4), resulting in a lack of unifying comparison (c.f., Section 5). This table does not aim at serving to rank or be a qualitatively comparison of surveyed methods, and one has to exercise caution when

comparing different setups. As a systematic organization of knowledge, we believe the table may be useful to practitioners looking for methods that satisfy certain properties, and useful for researchers who want to identify open problems and methods to further develop. Offering recourse in diverse settings with desirable properties remains an open challenge, which we explore in the following sections.

3 FORMULATION

Given a fixed predictive model, commonly assumed to be a binary classifier, $h : \mathcal{X} \rightarrow \{0, 1\}$, with $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_D$, we can define the set of *contrastive explanations* for a (factual) input $\mathbf{x}^F \in \mathcal{X}$ as $\mathcal{E} := \{\mathbf{x}^{CF} \in \mathcal{P}(\mathcal{X}) \mid h(\mathbf{x}^{CF}) \neq h(\mathbf{x}^F)\}$. Here, $\mathcal{P}(\mathcal{X}) \subseteq \mathcal{X}$ is a *plausible* subspace of $\mathcal{X}$, according to the distribution of training data (see Section 3.3.1). Descriptively, contrastive explanations identify alternative feature combinations (in nearby worlds [136]) that result in a favorable prediction from the fixed model. Assuming a notion of dissimilarity between instances, represented as $\text{dist}(\cdot, \cdot) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_+$, one can identify *nearest contrastive explanations* (a.k.a. counterfactual explanations) as follows:

$$\begin{aligned} \mathbf{x}^{*CF} \in \underset{\mathbf{x}^{CF} \in \mathcal{P}(\mathcal{X})}{\text{argmin}} \quad & \text{dist}(\mathbf{x}^{CF}, \mathbf{x}^F) \\ \text{s.t.} \quad & h(\mathbf{x}^{CF}) \neq h(\mathbf{x}^F) \\ & \mathbf{x}^{CF} = \mathbf{x}^F + \delta, \end{aligned} \tag{1}$$

where δ is the perturbation applied independently to the feature vector $\mathbf{x}^F$ to obtain the counterfactual instance $\mathbf{x}^{CF}$ [234]. As discussed in Section 2, although contrastive explanations identify the *feature vectors* that would achieve recourse, in general, the *set of actions* that would need to be performed to realize these features are not directly implied from the explanation [114]. Thus, a *consequential recommendation* for (factual) input $\mathbf{x}^F \in \mathcal{X}$ is defined as $\mathcal{R} := \{a \in \mathcal{A}(\mathbf{x}^F) \mid h(\mathbf{x}^{SCF}(a; \mathbf{x}^F)) \neq h(\mathbf{x}^F)\}$. Here, $\mathcal{A}(\mathbf{x}^F)$ is the set of *feasible* actions that can be performed by the individual seeking recourse (see Section 3.3.2). Approaching the recourse problem from a causal perspective within the SCM framework [114], actions are considered as interventions of the form $a = \text{do}(\{X_i := x_i^F + \theta_i\}_{i \in I}) \in \mathcal{A}(\mathbf{x}^F)$, and $\mathbf{x}^{SCF}(a; \mathbf{x}^F)$ denotes the structural counterfactual of $\mathbf{x}^F$ had action a been performed, all else being equal [184]. Finally, given a notion of cost of actions, capturing the effort expended by the individual as $\text{cost}(\cdot; \cdot) : \mathcal{A} \times \mathcal{X} \rightarrow \mathbb{R}_+$, one can identify *minimal consequential recommendations* as follows:

$$\begin{aligned} a^* \in \underset{a \in \mathcal{A}(\mathbf{x}^F)}{\text{argmin}} \quad & \text{cost}(a; \mathbf{x}^F) \\ \text{s.t.} \quad & h(\mathbf{x}^{CF}) \neq h(\mathbf{x}^F) \\ & \mathbf{x}^{CF} = \mathbf{x}^{SCF}(a; \mathbf{x}^F). \end{aligned} \tag{2}$$

Importantly, the solution of (1) yields a nearest contrastive explanation (i.e., $\mathbf{x}^{CF} = \mathbf{x}^F + \delta^*$), with no direct mapping to a set of (minimal) consequential recommendations [114]. Conversely, solving (2) yields both a minimal consequential recommendation (i.e., a^*) and a contrastive explanation (i.e., by construction $\mathbf{x}^{CF} = \mathbf{x}^{SCF}(a^*; \mathbf{x}^F)$).³

Our position is that, with the aim of providing recourse, the primary goal should be to provide minimal consequential recommendations that result in a (not necessarily nearest) contrastive

³Relatedly, the counterfactual instance that results from performing optimal actions, a^* , need not correspond to the counterfactual instance resulting from optimally and independently shifting features according to δ^* ; see Proposition 4.1 and Figure 1 in the work of Karimi et al. [114]. This discrepancy may arise due to, for example, minimal recommendations suggesting that actions be performed on an ancestor of those variables that are input to the model.

explanation when acted upon. Offering nearest contrastive explanations that are not necessarily attainable through *minimal* effort is of secondary importance to the individual. In practice, however, due to the additional assumptions needed to solve (2) (specifically for computing $\mathbf{x}^{\text{SCF}}$), the literature often resorts to solving (1). Despite the prevalence of methods assuming independent features, we quote the negative result of Karimi et al. [115, prop. 2], whereby the authors show that without full specification of the causal relations among variables, recourse cannot be guaranteed. As pointed out by Chou et al. [47], explanations derived from popular algorithms based on non-causal spurious correlations “may yield sub-optimal, erroneous, or even biased explanations.” Therefore, for the rest of this section, we operate on the assumption of knowledge of an SCM, which allows for a deterministic formulation of recourse yielding both minimal consequential recommendation and resulting contrastive explanations. Later in Sections 5 and 6, we relax these assumptions and discuss the relatively easier settings in which causal assumptions are scarce or absent entire. This is not to say that counterfactual explanations not accompanied by recommendation actions are not useful. On the contrary, as noted by Karimi et al. [114], counterfactual explanations hold promise for “guided audit of the data” [234] and evaluating various desirable model properties, such as robustness [98, 209] or fairness [96, 113, 209, 223] (see also Section 5). Besides this, it has been shown that designers of interpretable ML systems use counterfactual explanations for predicting model behavior [125], uncovering inaccuracies in the data profile of individuals [226], and also for medical outcome prediction under hypothesized treatments [153] or realistic medical image generation [238]. Beyond the consequential domains mentioned earlier and in Section 1, counterfactual explanations have been used for changing behavior of source code [48], overcoming class imbalance issues by amending underrepresented classes with plausible counterfactuals [217], and aiding the interpretable design of molecules [239].

In the rest of this section, we provide an overview of the objective function and constraints used in (1) and (2), followed by a description of the datatypes commonly used in recourse settings. Finally, we conclude with related formulations. Then in Section 4, we review the tools used to solve the formulations defined here.

3.1 Optimization Objective

Generally, it is difficult to define dissimilarity (*dist*) between individuals or cost functions for effort expended by individuals. Notably, this challenge was first discussed in the algorithmic fairness literature [73, 107] and later echoed throughout the algorithmic recourse community [21, 226]. In fact, “the law provides no formal guidance as to the proper metric for determining what reasons are most salient” [208]. Despite this, existing works have presented various ad hoc formulations with sensible intuitive justifications or practical allowance, which we review in the following.

3.1.1 On *dist*. Wachter et al. [234] define *dist* as the Manhattan distance, weighted by the inverse **median absolute deviation (MAD)**:

$$\text{dist}(\mathbf{x}, \mathbf{x}^{\text{F}}) = \sum_{k \in [D]} \frac{|\mathbf{x}_k - \mathbf{x}_k^{\text{F}}|}{\text{MAD}_k} \quad (3)$$

$$\text{MAD}_k = \text{median}_{j \in P} (|X_{j,k} - \text{median}_{l \in P}(X_{l,k})|).$$

This distance has several desirable properties, including accounting and correcting for the different ranges across features through the MAD heuristic, robustness to outliers with the use of the median absolute difference, and, finally, favoring sparse solutions through the use of ℓ_1 Manhattan distance.

Karimi et al. [113] propose a weighted combination of ℓ_p norms as a flexible measure across a variety of situations. The weights, $\alpha, \beta, \gamma, \zeta$ as shown in the following, allow practitioners to

balance between sparsity of changes (i.e., through the ℓ_0 norm), an elastic measure of distance (i.e., through the ℓ_1, ℓ_2 norms) [63], and a maximum normalized change across all features (i.e., through the ℓ_∞ norm):

$$\text{dist}(\mathbf{x}; \mathbf{x}^F) = \alpha \|\delta\|_0 + \beta \|\delta\|_1 + \gamma \|\delta\|_2 + \zeta \|\delta\|_\infty, \quad (4)$$

where $\delta = [\delta_1, \dots, \delta_D]^T$ and $\delta_k : \mathcal{X}_k \times \mathcal{X}_k \rightarrow [0, 1] \forall k \in [D]$. This measure accounts for the variability across heterogeneous features (see Section 3.5) by independently normalizing the change in each dimension according to its spread. Additional weights may also be used to relative emphasize changes in specific variables. Finally, other works aim to minimize dissimilarity on a graph manifold [189], in terms of Euclidean distance in a learned feature space [109, 183], or using a Riemannian metric in a latent space [14, 15].

3.1.2 On cost. Similar to Karimi et al. [113], various works explore ℓ_p norms to measure cost of actions. Ramakrishnan et al. [190] explore ℓ_1, ℓ_2 norm as well as constant cost if specific actions are undertaken; Karimi et al. [114, 115] minimize the ℓ_2 norm between $\mathbf{x}^F$ and the action $\mathbf{a}$ assignment (i.e., $\|\theta\|_2$); and Cui et al. [55] explore combinations of ℓ_0, ℓ_1, ℓ_2 norms over a user-specified cost matrix. Encoding individual-dependent restrictions is critical—for example, obtaining CREDIT is more difficult for an foreign student than for a local resident.

Beyond ℓ_p norms, the work of Ustun et al. [223] propose the total- and maximum-log percentile shift measures, to automatically account for the distribution of points in the dataset—for example,

$$\text{cost}(\mathbf{a}; \mathbf{x}^F) = \max_{k \in [D]} |Q_k(\mathbf{x}_k^F + \theta_k) - Q_k(\mathbf{x}_k^F)|, \quad (5)$$

where $Q_k(\cdot)$ is the CDF of x_k in the target population. This type of metric naturally accounts for the relative difficulty of moving to unlikely (high or low percentile) regions of the data distribution. For instance, going from a 50th to a 55th percentile in SCHOOL GRADES is simpler than going from the 90th to the 95th percentile.

3.1.3 On the Relation Between dist and cost. In a world in which changing one variable does not affect others, one can see a parallel between the counterfactual instance of (1) (i.e., $\mathbf{x}^{\text{CF}} = \mathbf{x}^F + \delta$) and that of (2) (i.e., $\mathbf{x}^{\text{CF}} = \mathbf{x}^{\text{SCF}}(\mathbf{a}; \mathbf{x}^F) = \mathbf{x}^F + \theta$). This mirroring form suggests that definitions of dissimilarity between individuals (i.e., dist) and effort expended by an individual (i.e., cost) may be used interchangeably. Following Karimi et al. [114], however, we do not consider a general 1-1 mapping between dist and cost. For instance, in a two-variable system with MEDICATION as the parent of HEADACHE, an individual who consumes more than the recommended amount of medication may not recover from the headache (i.e., higher cost but smaller *symptomatic* distance relative to another individual who consumed the correct amount). Furthermore, although dissimilarity is often considered to be symmetric (i.e., $\text{dist}(\mathbf{x}_A^F, \mathbf{x}_B^F) = \text{dist}(\mathbf{x}_B^F, \mathbf{x}_A^F)$), the effort needed to go from one profile to another need not satisfy symmetry—for example, spending money is easier than saving money (i.e., $\text{cost}(\mathbf{a} = \text{do}(X_{\$} := \mathbf{x}_{A,\$}^F - \$500); \mathbf{x}_A^F) \leq \text{cost}(\mathbf{a} = \text{do}(X_{\$} := \mathbf{x}_{B,\$}^F + \$500); \mathbf{x}_B^F)$). These examples illustrate that the interdisciplinary community must continue to engage to define the distinct notions of dist and cost, and such definitions cannot arise from a technical perspective alone.

3.2 Model and Counterfactual Constraints

3.2.1 Model. A variety of fixed models have been explored in the literature for which recourse is to be generated. As summarized in Table 1, we broadly divide them in four categories: tree-based (TB), kernel-based (KB), differentiable (DF), and other (OT) types (e.g., generalized linear models, naive Bayes, k -nearest neighbors). Although the literature on recourse has primarily focused on binary classification settings, most formulations can easily be extended to multi-class classification

or regression settings. Extensions to such settings are straightforward, where the model constraint is replaced with $h(\mathbf{x}^{\text{CF}}) = k$ for a target class or $h(\mathbf{x}^{\text{CF}}) \in [a, b]$ for a desired regression interval, respectively. Alternatively, soft predictions may be used in place of hard predictions, where the goal may be, for example, to increase the prediction gap between the highest-predicted and second-highest-predicted class—that is, $\text{Pred}(\mathbf{x}^{\text{CF}})_i - \text{Pred}(\mathbf{x}^{\text{CF}})_j$, where $i = \text{argmax}_{k \in K} \text{Pred}(\mathbf{x}^{\text{CF}})_k$, $j = \text{argmax}_{k \in K \setminus i} \text{Pred}(\mathbf{x}^{\text{CF}})_k$. In the end, the model constraint representing the change in prediction may be arbitrarily non-linear, non-differentiable, and non-monotone [21], which may limit the applicability of solutions (c.f. Section 4).

3.2.2 Counterfactual. The counterfactual constraint depends on the type of recourse offered. Whereas $\mathbf{x}^{\text{CF}} = \mathbf{x}^{\text{F}} + \delta$ in (1) is a linear constraint, computing $\mathbf{x}^{\text{CF}} = \mathbf{x}^{\text{SCF}}(\mathbf{a}; \mathbf{x}^{\text{F}})$ in (2) involves performing the three step abduction-action-prediction process of Pearl et al. [186] and may thus be non-parametric and arbitrary involved. A closed-form expression for deterministically computing the counterfactual in additive noise models is presented in the work of Karimi et al. [114], and probabilistic derivations for more general SCMs are presented in another work by Karimi et al. [115].

3.3 Actionability and Plausibility Constraints

3.3.1 Plausibility. Existing literature has formalized plausibility constraint as one of three categories: *domain-consistency*, *density-consistency*, and *prototypical-consistency*. Whereas domain-consistency restricts the counterfactual instance to the range of admissible values for the domain of features [113], density-consistency focuses on likely states in the (empirical) distribution of features [63, 64, 109, 111, 131, 183] identifying instances close to the data manifold. A third class of plausibility constraints selects counterfactual instances that are either directly present in the dataset [189, 241] or close to a prototypical example of the dataset [11, 12, 124, 131, 225].

3.3.2 Actionability (Feasibility). The set of feasible actions, $\mathcal{A}(\mathbf{x}^{\text{F}})$, is the set of interventions $\text{do}(\{X_i := x_i^{\text{F}} + \theta_i\}_{i \in I})$ that the individual, $\mathbf{x}^{\text{F}}$, is able to perform. To determine $\mathcal{A}(\mathbf{x}^{\text{F}})$, we must identify the set of variables upon which interventions are possible, as well as the pre-/post-conditions that the intervention must satisfy. The actionability of each variable falls into three categories [114, 130]:

- (I) Actionable (and mutable) (e.g., BANK BALANCE);
- (II) Mutable but non-actionable (e.g., CREDIT SCORE);
- (III) Immutable (and non-actionable) (e.g., BIRTHPLACE).

Intuitively, mutable but non-actionable variables are not directly actionable by the individual but may change as a consequence of a change to its causal ancestors (e.g., REGULAR DEBT PAYMENT).

Having identified the set of actionable variables, an intervention can change the value of a variable *unconditionally* (e.g., BANK BALANCE can increase or decrease), or *conditionally* to a specific value [114] or in a specific direction [223]. Karimi et al. [114] present the following examples to show that the actionable feasibility of an intervention on X_i may be contingent on any number of conditions:

- (i) Pre-intervention value of the intervened variable (i.e., x_i^{F})—for example, an individual’s AGE can only increase (i.e., $[x_{\text{AGE}}^{\text{SCF}} \geq x_{\text{AGE}}^{\text{F}}]$);
- (ii) Pre-intervention value of other variables (i.e., $\{x_j^{\text{F}}\}_{j \in [d] \setminus i}$)—for example, an individual cannot apply for CREDIT on a temporary VISA (i.e., $[x_{\text{VISA}}^{\text{F}} = \text{PERMANENT}] \geq [x_{\text{CREDIT}}^{\text{SCF}} = \text{TRUE}]$);

- (iii) Post-intervention value of the intervened variable (i.e., x_i^{SCF})—for example, an individual may undergo heart surgery (an additive intervention) only if they will not remiss due to sustained SMOKING HABITS (i.e., $[x_{\text{HEART}}^{\text{SCF}} \neq \text{REMISSION}]$);
- (iv) Post-intervention value of other variables (i.e., $\{x_j^{\text{SCF}}\}_{j \in [d] \setminus i}$)—for example, an individual may undergo heart surgery only *after* their blood pressure (BP) is regularized due to medicinal intervention (i.e., $[x_{\text{BP}}^{\text{SCF}} = \text{O.K.}] \geq [x_{\text{HEART}}^{\text{SCF}} = \text{SURGERY}]$).

All such feasibility conditions can easily be encoded as Boolean/logical constraint into $\mathcal{A}(\mathbf{x}^F)$ and jointly solved for in the constrained optimization formulations (1), (2). An important side note to consider is that $\mathcal{A}(\mathbf{x}^F)$ is *not* restricted by the SCM assumptions but instead by individual-/context-dependent consideration that determine the *form*, *feasibility*, and *scope* of actions [114].

3.3.3 On the Relation Between Actionability and Plausibility. While seemingly overlapping, *actionability* (i.e., $\mathcal{A}(\mathbf{x}^F)$) and *plausibility* (i.e., $\mathcal{P}(X)$) are two distinct concepts: whereas the former restrict actions to those that are *possible to do*, the latter require that the resulting counterfactual instance be *possibly true*, *believable*, or *realistic*. Consider a Middle Eastern Ph.D. student who is denied a U.S. visa to attend a conference. Although it is quite likely for there to be favorably treated foreign students from other countries with similar characteristics (*plausible* GENDER, FIELD OF STUDY, ACADEMIC RECORD, etc.), it is impossible for our student to act on their BIRTHPLACE for recourse (i.e., a *plausible* explanation but an *infeasible* recommendation). Conversely, an individual may perform a set of *feasible* actions that would put them in an *implausible* state (e.g., small $p(\mathbf{x}^{\text{CF}})$; not in the dataset) where the model fails to classify with high confidence. Thus, actionability and plausibility constraints may be used in conjunction depending on the recourse setting they describe.

3.4 Diversity and Sparsity Constraints

3.4.1 Diversity. Diverse recourse is often sought in the presence of uncertainty (e.g., unknown user preferences when defining *dist* and *cost*). Approaches seeking to generate diverse recourse generally fall in two categories: diversity through multiple runs of the same formulation or diversity via explicitly appending diversity constraints to prior formulations.

In the first camp, Wachter et al. [234] show that different runs of their gradient-based optimizer over a non-convex model (e.g., multilayer perceptron) results in different solutions as a result of different random seeds. Sharma et al. [209] show that multiple evolved instances of the genetic-based optimization approach can be used as diverse explanations, hence benefiting from not requiring multiple re-runs of the optimizer. Downs et al. [69], Mahajan et al. [146], and Pawelczyk et al. [183] generate diverse counterfactuals by passing multiple samples from a latent space that is shared between factual and counterfactual instances through a decoder, and filtering those instances that correctly flip the prediction.

In the second camp, Russell [205] pursues a strategy whereby subsequent runs of the optimizer would prevent changing features in the same manner as prior explanations/recommendations. Karimi et al. [113] continue in this direction and suggest that subsequent recourse should not fall within an ℓ_p -ball surrounding any of the earlier explanations/recommendations. Cheng et al. [46] and Mothilal et al. [167] present a differentiable constraint that maximizes diversity among generated explanations by maximizing the determinant of a (kernel) matrix of the generated counterfactuals.

3.4.2 Sparsity. It is often argued that sparser solutions are desirable as they emphasize fewer changes (in explanations) or fewer variables to act upon (in recommendations) and are thus more interpretable for the individual [154]. Although this is not generally accepted [182, 225], one

can formulate this requirement as an additional constraint, whereby, for example, $\|\delta\|_0 \leq s$ or $\|\theta\|_0 \leq s$. Formulating sparsity as an additional (hard) constraint rather than optimizing for it in the objective grants the flexibility to optimize for a different object while ensuring that a solution would be sparse.

3.5 Datatypes and Encoding

A common theme in consequential decision-making settings is the use of datatypes that refer to real-world attributes of individuals. As a result, datasets are often tabular , comprising heterogeneous features, with a mix of numeric (real, integer), binary, categorical, or ordinal variables. Most commonly, Adult [3], Australian Credit [71], German Credit [17], GiveMeCredit [248], COMPAS [127], and HELOC [103], among others, are used, which are highly heterogeneous.

Different feature types obey different statistical properties—for example, the integer-based HEART RATE, real-valued BMI, categorical BLOOD TYPE, and ordinal AGE GROUP differ drastically in their range. Thus, heterogeneous data requires special handling to preserve their semantics. A common approach is to encode each variable according to a predetermined strategy, which pre-processes the data before model training and consequently during recourse generation (for either recourse type, i.e., consequential recommendations or contrastive explanations). For instance, categorical and ordinal features may be encoded using one-hot encoding and thermometer encoding, respectively. To preserve the semantics of each variable during recourse generation, we must also ensure that the generated explanations/recommendations result in counterfactual instances that also satisfy the encoding constraints. For instance, Boolean and linear constraints of the form $\sum_j x_{i,j} = 1 \forall x_{i,j} \in \{0, 1\}$ are used to ensure that multiple categories are not simultaneously active, and thermometer-encoded ordinal variables are required to satisfy $x_{i,j} \geq x_{i,j+1} \forall x_{i,j} \in \{0, 1\}$. For a detailed overview, refer to the work of Nazabal et al. [171].

In addition to  tabular data, one may require contrastive explanations for  image-based or  text-based datasets, as summarized in Table 1. For image-based datasets, the algorithm may optionally operate on the raw data, or on super-pixel or other forms of extracted features (e.g., a hidden representation in a neural network). Text-based datasets are also commonly encoded as vectors representing GloVe [187] or bag-of-words embeddings [198]. For a recent work that explores the use of counterfactual explanations for functional data, refer to Carrizosa et al. [42].

3.6 Related Formulations

The problem formulation for recourse generation, and specifically that of contrastive explanations, (1), is broadly related to several other problems in data mining and ML. For instance, the *cost-minimizing inverse classification problem* [4, 128–130, 132, 147] aims to identify the “minimum required change to a case in order to reclassify it as a member of a different preferred class” [147]. *Actionable knowledge extraction* is employed in data mining to suggest “behaviors which render a state of an instance into a preferred state” [70] according to a classifier [37–39, 112, 247]. Finally, *adversarial perturbations* are small imperceptible changes to the input of a classifier that would alter the output prediction to a false and highly confident region [40, 88, 164, 165, 172, 177, 178, 216]. An additional parallel shared by the preceding methods is in their assumption of a fixed underlying model. Extensions of the preceding, in which model designers anticipate and aim to prevent malicious behavior, exist in the *strategic classification* [67, 99, 104, 120, 135, 140, 155, 158] and *adversarial robustness* [41, 49, 76, 246] literature. For recent expositions on the similarities and contrasts between these methods, refer to works elsewhere [75, 78, 180].

Whereas there exists strong parallels in their formulations, the differences arise in their intended use cases and guarantees for the stakeholders involved. For example, as opposed to recourse that aims to build trust with affected individuals, the primary use case cited in the actionable

knowledge extraction literature is to deliver cost-effective actions to maximize profit or other business objectives. Furthermore, whereas a contrastive explanation aims to inform an individual about ways in which their situation would have led to a desirable outcome, an adversarial perturbation aims to fool the human by being imperceptible (e.g., by leaving the data distribution). In a sense, imperceptibility is the antithesis of explainability and trust. Finally, building on the presentation in Section 2, offering consequential recommendations relies on a causal modeling of the world, which is largely ignored by other approaches.

4 SOLUTION

By definition, recourse is offered when an individual is presented with contrastive explanations and consequential recommendations, which can be obtained by solving (1) and (2), respectively. Notably, the objective that is to be minimized (i.e., dist or cost) may be non-linear, non-convex, or non-differentiable. Furthermore, without restricting the classifier family, the model constraint also need not be linear, monotonic, or convex. Finally, based on individual-/context-specific restrictions, the problem setting may require optimizing over a constrained set of plausible instances, $\mathcal{P}(\mathcal{X})$, or feasible actions, $\mathcal{A}(\mathbf{x}^F)$.⁴ Thus, a distance/cost-agnostic and model-agnostic solution with support for plausibility, feasibility, sparsity, and diversity constraints over heterogeneous datasets will in general require complex approaches, trading-off various desirable properties in the process. In the following, we discuss the importance of these properties and provide an overview of utilized solutions.

4.1 Properties

We remark that the optimizer and the resulting solutions should ideally satisfy some desirable properties, as detailed next. In practice, methods typically trade off optimal guarantee δ^* , perfect coverage Ω^* , or efficient runtime τ^* , and may otherwise require prohibitive *access* to the underlying data or predictive model.

4.1.1 Optimality. Identified counterfactual instances should ideally be *proximal* to the factual instance, corresponding to a small change to the individual's situation. When optimizing for minimal dist and cost in (1) and (2), the objective functions and constraints determine the *existence* and *multiplicity* of recourse. For factual instance $\mathbf{x}^F$, there may exist zero, one, or multiple⁵ optimal solutions, and an ideal optimizer should thus identify (at least) one solution (explanation or recommendation, respectively) if one existed, or terminate and return N/A otherwise.

4.1.2 Perfect Coverage. Coverage is defined as the number of individuals for which the algorithm can identify a plausible counterfactual instance (through either recourse type), if at least one solution existed [113]. Communicating the domain of applicability to users is critical for building trust [161, 234], and ideally the algorithm offers recourse to all individuals (i.e., perfect coverage).

4.1.3 Efficient Runtime. As explanations/recommendations are likely to be offered in conversational settings [101, 105, 157, 212, 240], it is desirable to generate recourse in near-real time. Despite the general underreporting of runtime complexities in the literature, in Table 1 we highlight those algorithms that report efficient wall-clock runtimes that may enable interaction between the algorithm and individual.

4.1.4 Access. Different optimization approaches may require various levels of access to the underlying dataset or model. Access to the model may involve *query access* (where only the label is

⁴Optimization terminology refers to both of these constraint sets as *feasibility* sets.

⁵The existence of multiple equally costly recourse actions is commonly referred to as the Rashoman effect [34].

returned), *gradient access* (where the gradient of the output with respect to the input is requested), *class probabilities access* (from which one can infer the confidence of the prediction), or complete *white-box access* (where all model parameters are known).

Naturally, there are practical implications to how much access is permissible in each setting, which further restricts the choice of tools. Consider an organization that seeks to generate recourse for their clients. Unless these algorithms are run in-house by the said organization, it is unlikely that the organization would hand over training data, model parameters, or even non-rate-limited API of their models to a third-party to generate recourse.

4.2 Tools

We consider the richly explored field of optimization [33, 174, 214] outside the scope of this work and suffice to briefly review the tools used specifically for recourse generation, highlighting their domain of applicability, and relegating technical details to appropriate references. Not only is solving (1) and (2) difficult in general settings [130], it has even been shown to be NP-complete or NP-hard in restricted settings, such as solving for integer-based variables [12], solving for additive tree models [16, 55, 219] or neural networks [116], and solving for quadratic objectives and constraints [12, 33, 179]. Thus, except for exhaustive search over a potentially uncountable set of solutions, most works pursue *approximate* solutions in restricted settings, trading off the preceding desirable properties (see Table 1). Solutions can be broadly categorized as *gradient-based optimization*, *model-based*, *search-based*, *verification-based*, and *heuristics-based*.⁶

Under differentiability of the objective and constraints, gradient-based optimization solutions such as FISTA [26] are employed [63, 64, 225] to find globally optimal solutions under convex Lagrangian, and first-order methods such as (L-)BFGS or projected gradient descent may be used to identify local optima otherwise. Relatedly, rather than solving recourse for each individual independently, some works pursue a model-based approach, whereby a mapping from factual to counterfactual instances is learned through gradient optimization [146, 183]. These methods enjoy efficient runtimes at the cost of coverage loss and poor handling of heterogeneous data.

For non-differentiable settings, branch-and-bound-based [134] approaches split the *search* domain into smaller regions within which a solution may be easier to find. Under linearity of the objectives and constraints, **integer linear programming (ILP)** algorithms may be used when datatypes are discrete [55, 223], and **mixed-integer linear programming (MILP)** extensions are utilized when some variables are not discrete [110, 205]. (M)ILP formulations are solved using powerful off-the-shelf solvers such as CPLEX [54] and Gurobi [176]. One may also use a combination of iterative binary *search* and *verification* tools to obtain solutions to (1) and (2). Here, the problem is reformulated as a constrained satisfaction problem, where the constraint corresponding to the objective (dist or cost) is updated in each iteration to reflect the bounds in which a solution is obtainable [113, 114, 162]. As with (M)ILP, this approach benefits from the existence of off-the-shelf solvers such as Z3 [57], CVC4 [23], and pySMT [81]. The problem may also be cast and solved as program synthesis [59, 190] or answer-set programming [29]. The preceding methods typically offer optimality and perfect coverage while relying on white-box access to the fixed model parameters.

A number of heuristics-based approaches are also explored, such as finding the shortest path (Dijkstra's algorithm [53]) between $\mathbf{x}^F$ and potential $\mathbf{x}^{CF}$ s on an empirical graph where edges are placed between similar instances (according to, e.g., Gaussian kernel) [189]. Finally, genetic-based approaches [243, 252] find solutions over different evolutions of candidate solutions according to

⁶Alternative categorization of recourse generating methods can be found in the work of Redelmeier et al. [196].

various heuristics [22, 56, 93, 124, 209] and benefit from being model-/datatype-/norm-agnostic via only requiring query access to the model.

5 PROSPECTS

In the previous sections, we covered the definitions, formulations, and solutions of existing works aiming to offer algorithmic recourse. We showed that generating recourse explanations and recommendations required counterfactual reasoning based on different levels of causal knowledge. Counterfactual reasoning has roots not only in the philosophy of science [101, 102, 136–138, 244] but also in the psychology of human agents [36, 156, 157] and benefits from strong technical foundations [19, 97]. User studies have demonstrated that causal relationships are assessed by evaluating counterfactuals [151], and counterfactual simulation is used to predict future events [83]. Specifically in the context of XAI, it has been shown that counterfactuals can “make the decisions of inscrutable systems intelligible to developers and users” [36], and that people perform better at predicting model behavior when presented with counterfactual instances [125]. Organizations seek to deploy counterfactual-based explanations citing their easy-to-understand nature [30, 31] and GDPR compliance [234]. Finally, from a practitioner’s standpoint, not only does algorithmic recourse benefit from the widely exercised practice of sharing open source implementations (see Table 1), but various graphical interfaces have also been developed to assist the onboarding of non-technical stakeholders [46, 87, 241].

There are, however, several implicit assumptions made in existing setups—for example, that the world dynamics are known and do not change, that the predictive (supervised) model is fixed, and that changes only arise due to the actions of the individual seeking recourse. Moreover, in the multi-agent settings considered (with, e.g., banks and loan seekers), agents are assumed to act truthfully with no gaming or false reporting of features, and agents are aligned in the aim to minimize an agreed-upon objective function. In the following, we explore settings in which these assumptions do not hold and offer potential solutions for extending to more realistic recourse settings.

5.1 Beyond Deterministic Recourse

In (2), we saw that minimal consequential recommendations are generated subject to the constraint that the counterfactual instance, $\mathbf{x}^{\text{CF}}$, is assigned to be the structural counterfactual of $\mathbf{x}^{\text{F}}$ under hypothetical actions $\mathbf{a}$ —that is, $\mathbf{x}^{\text{CF}} = \mathbf{x}^{\text{SCF}}(\mathbf{a}; \mathbf{x}^{\text{F}})$ [114]. Computing the structural counterfactual exactly, however, relies on strong assumptions (i.e., the true SCM is an additive noise model and is known). Karimi et al. [115] show that without complete knowledge of the true SCM, counterfactual analysis cannot be done exactly and thus recourse cannot be offered deterministically. Although they then present methods that offer recourse with high probability, they do so under specification of a causally sufficient graph. Future research in this direction may explore less strict settings, perhaps accounting for hidden confounders [232], or partially observed graphs [9, 51, 218], further adding to the uncertainty of recourse recommendations [27, 122]. Alternatively, sources of stochasticity may enter the recourse process via a non-deterministic decision-making system. For example, it has been demonstrated that for models trained on *selective labels*, fair and optimal decisions should be made stochastically [25, 119, 221].

5.2 Beyond Supervised Recourse

In Section 3.2 we discussed how the standard binary classification setting could be extended to support multi-class classification and regression. Beyond these classic supervised learning settings, an individual may be subject to an automated decision maker that determines a matching of applicants to resources across a population (e.g., kindergarten assignment for children or housing for low-income families). Alternatively, one can expect to generate explanations in more interactive

settings, such as for the actions and policies of a reinforcement learning agent [144, 145, 201, 224] or for recommender systems [60, 84]. Finally, explanations may also be generated for time-series data [5, 16, 142], which can be extended to support online data streams and models that change over time [21, 61, 182, 200, 226].

5.3 Beyond Individualized Recourse

So far, the presented formulations aimed to offer recourse explanations pertaining to a single individual and assumed that recourse recommendations would be undertaken by that individual. However, it is natural to extend the notion of recourse beyond the data subject in question, or beyond a single individual in the population.

An example of the former setting is when the family member of a patient decides on a treatment on their behalf when the patient cannot directly exercise their agency due to incapacitation [226]. One may also consider common cases in judicial processes where a legal counsel represents and seeks recourse for their client that may then be exercised by another fiduciary. In such settings, the formulation of cost and feasibility of actions may need to be adjusted to account for restrictions on both the subject and the actor.

Alternatively, recourse may be achieved through the collective action of a group of people rather than that of a single individual [114]. For instance, the efforts of social and political activists may culminate in a LAW change that offers better conditions for a group of individuals. In such settings, a (background) variable that is non-actionable (or incurs high cost) on an individual level may be rendered as actionable on a group level, which may in turn bring down the cost for all members of the group. This example also suggests that background variables may capture contextual information (e.g., ECONOMY) that are not characteristics of, but nonetheless affect, the individual. Furthermore, the individual may not have control over these macro variables that change over time and violate the stationarity assumption of the world. Additionally, explicit consideration for multiple agents [175], building theories of minds for each agent so as to best persuade [123], and viewing recourse as a sequential game under non-instantaneous effects of actions taken place [170, 228] is an underexplored area for future research. Finally, the need to analyze recourse on a subpopulation level may arise due to uncertainty in assumptions [115] or as an intentional study of other properties of the system, such as fairness [46, 96, 113, 195, 223], which we explore further in the following.

5.4 On the Interplay of Recourse and Ethical ML

The preceding research questions have primarily focused on one stakeholder: the affected individual. However, giving the right of recourse to individuals should not be considered in a vacuum and independently of the effect that providing explanations/recommendations may have on other *stakeholders* (e.g., model deployer and regulators), or in relation to *other desirable properties* (e.g., fairness, security, privacy, robustness), broadly referred to as ethical ML. We explore this interplay next.

5.4.1 Recourse and Fairness. Fairness in ML is a primary area of study for researchers concerned with uncovering and correcting potentially discriminatory behavior of ML models. In this regard, prior work has informally used the concept of *fairness of recourse* as a means to evaluate the *fairness of predictions*. For instance, Ustun et al. [223] look at comparable male/female individuals that were denied a loan and show that a disparity can be detected if the suggested recourse actions (namely, *flipsets*) require relatively more effort for individuals of a particular subgroup. Along these lines, Sharma et al. [209] evaluate group fairness via aggregating and comparing the cost of recourse (namely, *burden*) over individuals of different subpopulations. Karimi et al. [113] show that the

addition of feasibility constraints (e.g., non-decreasing AGE) that results in an increase in the cost of recourse indicates a reliance of the fixed model on the sensitive attribute AGE, which is often considered as legally and socially unfair. Here we clarify that these notions are distinct and would benefit from a proper mathematical study of the relation between them.

The preceding examples suggest that evidence of discriminatory recourse (e.g., reliance on RACE) may be used to uncover unfair classification. We show, however, that the contrapositive statement does not hold: consider, for example, a 2D dataset comprising of two subgroups (i.e., $s \in \{0, 1\}$), where $p(x|s) = \mathcal{N}(0, 10^s)$. Consider a binary classifier, $h : \mathbb{R} \times \mathbb{S} \rightarrow \{0, 1\}$, where $h(x, s) = \text{sign}(x)$. Although the distribution of predictions satisfies demographic parity, the (average) recourse actions required of negatively predicted individuals in $s = 1$ is larger than those in $s = 0$. Thus, we observe unfair recourse even when the predictions are demographically fair.

This contradiction (the conditional and contrapositive not holding simultaneously) can be resolved by considering a new and distinct notion of fairness (i.e., *fairness of recourse*) that does not imply or is not implied by the *fairness of prediction*. In this regard, *equalizing recourse* was recently presented by Gupta et al. [96], who offered the first recourse-based (and prediction-independent) notion of fairness. The authors demonstrate that one can directly calibrate for the average distance to the decision boundary to be equalized across different subgroups during the training of both linear and non-linear classifiers. A natural extension would involve considering the *cost of recourse actions*, as opposed to the *distance to the decision boundary*, in flipping the prediction across subgroups. In summary, recourse may trigger new definitions of fairness to ensure non-discriminatory behavior of ML models, as in the work of Von Kügelgen et al. [233].

5.4.2 Recourse and Robustness. Robustness often refers to our expectation that model outputs should not change as a result of (small) changes to the input. In the context of recourse, we expect that similar individuals should receive similar explanations/recommendations, or that recourse suggestions for an individual should be to some extent invariant to the underlying decision-making system trained on the same dataset [204]. In practice, however, the stability of both gradient-based [8, 65, 85, 152] and counterfactual-based [131, 133] explanation systems has been called into question. Interestingly, it has been argued that it is possible for a model to have *robust predictions* but *non-robust explanations* [98], and vice versa [131] (similar to relation between fair predictions and fair recourse). Parallel studies argue that the sparsity of counterfactuals may contribute to non-robust recourse when evaluating explanations generated under different fixed models [182]. Finally, in the case of consequential recommendations, robustness will be affected by assumptions of the causal generative process (see Figure 1). Carefully reviewing assumptions and exploring such robustness issues in more detail is necessary to build trust in the recourse system and in turn in the algorithmic decision-making system. For recent works in this direction, which study the sensitivity of recourse methods to uncertainty in the features, model, or causal assumptions, refer to works found elsewhere [62, 66, 160, 173, 211, 222, 230]. Another interesting research question pertains to the generation of recourse when actionability of the recommended actions changes over time or due to external sources (e.g., inflation). Studying and accounting for various failure modes of recourse is important to offer robust and ultimately useful recourse.

5.4.3 Recourse, Security, and Privacy. Model extraction concerns have been raised in various settings for ML APIs [141, 197, 220, 236]. In such settings, an adversary aims to obtain a surrogate model, $\hat{f}$, that is similar (e.g., in fidelity) to the target model, f :

$$f \approx \hat{f} = \arg \min_{\hat{f} \in \mathcal{F}} \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} [\mathcal{L}(f(\mathbf{x}), \hat{f}(\mathbf{x}))]. \quad (6)$$

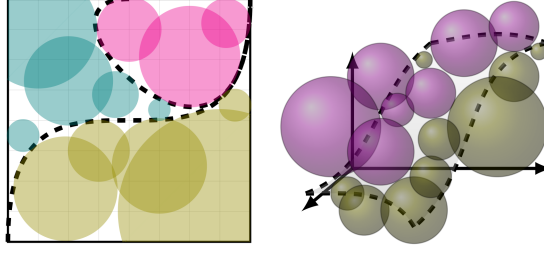


Fig. 2. Here we illustrate the model stealing process in 2D and 3D using hypothetical non-linear decision boundaries. “How many optimal contrastive explanations are needed to extract the decision regions of a classifier?” can be formulated as “How many factual balls are needed to maximally pack all decision regions?”

Here, an adversary may have access to various model outputs (classification label [141], class probabilities [220], etc.) under different query budgets (unlimited, rate-limited, etc. [44, 108]). Model extraction may be accelerated in the presence of additional information, such as gradients of outputs with respect to inputs⁷ [159], or contrastive explanations [7]. Related to recourse, and of practical concern [21, 208, 213, 223], is a study of the ability of an adversary with access to a recourse API in extracting a model. Specifically, we consider a setting in which the adversary has access to a *prediction API* and a *recourse API*, which given a factual instance, $\mathbf{x}^F$, returns a nearest contrastive explanation, $\mathbf{x}^{*CF}$, using a known distance function, $\Delta : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_+$.⁸ How many queries should be made to these APIs to perform a functionally equivalent extraction of $f(\cdot)$?

In a first attack strategy, one could learn a surrogate model on a dataset where factual instances and labels (from the training set or randomly sampled) are augmented with counterfactual instances and counterfactual labels. This idea was explored by Aïvodji et al. [7], where they demonstrated that a high-fidelity surrogate model can be extracted even under low query budgets. Although easy to understand and implement, this attack strategy implicitly assumes that constructed dataset has independent and identically distributed data, and thus does not make use of the relations between factual and counterfactual pairs.

An alternative attack strategy considers that the model f can be fully represented by its decision boundaries, or the complementary *decision regions* $\{\mathcal{R}_1, \dots, \mathcal{R}_l\}$. Every contrastive explanation returned from the recourse API informs us that all instances surrounding the factual instance, up to a distance of $\Delta(\mathbf{x}^F, \mathbf{x}^{*CF})$, share the same class label as $\mathbf{x}^F$ according to f (otherwise, that instance would be the nearest contrastive explanation). More formally, $f(\mathbf{x}^F) = f(\mathbf{x}) \forall \mathbf{x} \in \mathcal{B}_{\mathbf{x}^F}^\Delta$, where $\mathcal{B}_{\mathbf{x}^F}^\Delta$ is referred to as the Δ -factual ball, centered at $\mathbf{x}^F$ and with radius $\Delta(\mathbf{x}^F, \mathbf{x}^{*CF})$. The model extraction problem can thus be formulated as the number of factual balls needed to maximally pack all decision regions (Figure 2):

$$\Pr \left[\text{Vol}(\mathcal{R}_l) - \bigcup_{i=1, \mathcal{B}_{\mathbf{x}_i^F}^\Delta \subseteq \mathcal{R}_l}^n \text{Vol}(\mathcal{B}_{\mathbf{x}_i^F}^\Delta) \leq \epsilon \right] \geq 1 - \delta \quad \forall l. \quad (7)$$

As in other extraction settings, $\hat{f}$ can then be used to infer private information on individuals in the training set, to uncover exploitable system vulnerabilities, or for free internal use. Understanding

⁷A large class of explanation methods rely on the gradients to offer saliency/attribution maps, especially in the image domain.

⁸Explanation models such as MACE [113] provide optimal solutions, $\mathbf{x}_\epsilon^{CF}$, where $f(\mathbf{x}^F) \neq f(\mathbf{x}_\epsilon^{CF})$, $\Delta(\mathbf{x}^F, \mathbf{x}_\epsilon^{CF}) \leq \Delta(\mathbf{x}^F, \mathbf{x}^{*CF}) + \epsilon$, where $\mathbf{x}^{*CF}$ is the optimal nearest contrastive explanation. In practice, $\epsilon = 1e-5$, which in turn results in $\mathbf{x}_\epsilon^{CF} \approx \mathbf{x}^{*CF}$.

attack strategies may guide recourse policy and the design of defensive mechanisms to hinder the exploitation of such vulnerabilities.

Surprisingly, a model need not be extracted in the preceding sense to be revealing of sensitive information. Building on the preceding intuition, we note that a single contrastive explanation informs the data subject that there are no instances in a certain vicinity (i.e., within $\mathcal{B}_{x_f}^\Delta$) such that their prediction is different. This information informs the data subject about, for example, whether their similar friend was also denied a loan, violating their predictive privacy. Even under partial knowledge of the friend's attributes, an adversary may use the information about the shared predictions in $\mathcal{B}_{x_f}^\Delta$ to perform membership inference attacks [210] or infer missing attributes [72]. This problem is worsened when multiple diverse explanations are generated and is an open problem.

5.4.4 Recourse and Manipulation. Although a central goal of recourse is to foster trust between an individual and an automated system, it would be simplistic to assume that all parties will act truthfully in this process. For instance, having learned something about the decision-making process (perhaps through recommendations given to similar individuals), an individual may exaggerate some of their attributes for a better chance of favorable treatment [226]. Trust can also be violated by the recourse-offering party. As discussed earlier, the multiplicity of recourse explanations/recommendations (see Section 4.1.1) may allow for an organization to cherry-pick “the most socially acceptable explanation out of many equally plausible ones” [21, 98, 126] (also see *fairwashing* [6]). In such cases of misaligned incentives, the oversight of a regulatory body, perhaps with random audits of either party, seems necessary. Another solution may involve mandating a minimum number of diverse recourse offerings, which would conflict with security considerations.

5.5 Toward Unifying Benchmarks

Table 1 presents an overview of the diverse settings in which recourse is sought. Despite the abundance of open source implementations built on robust tools and working well in their respective settings, a comparative benchmark for recourse is lacking. This problem is exacerbated for consequential recommendations that further rely on assumptions about the causal generative process. To make objective progress, however, new contributions should be evaluated against existing methods. Thus, a next step for the community is the curation of an online challenge (e.g., using Kaggle) to benchmark the performance of existing and new methods. To broadly cover practical settings, we envision multiple tracks where authors can submit their generated explanations/recommendations given a fixed classifier, test dataset, and pre-determined dist/cost definition, and be evaluated using the metrics defined in Section 4.1. Authors may also submit results that satisfy additional actionability, plausibility, and diversity constraints, and be compared as such.⁹ Finally, besides striving for all-encompassing evaluation benchmarks for recourse, a push toward understanding the relations between recourse methods and other explainability methods is quite welcome. In particular, refer to other works [74, 77, 80, 121] for recent efforts that show how recourse can lead to, or result from, other such methods as attribution methods. User testing of recourse, given the consequential nature of the domain, is more difficult (if at all ethically possible) than explainability methods targeting other stakeholders in non-consequential domains. Nevertheless, with the ultimate objective of building not just explanations but reliable and robust explanations, future research should investigate ways that such uncertainties can be modeled (see Section 5.4.2). This is important not just for the recipient of the recourse recommendation but also for the

⁹Since the manuscript of this work was first published online, several works have pushed in this direction [58, 58, 106, 192, 251], and also a public repository for running many recourse generating methods in a unified and comparable fashion [181].

institution offering such recommendation, as, in a sense, it is a binding contract between the parties to offer some resource when certain conditions are met.

6 CONCLUSION

Our work started with a case study of a 28-year-old female professional who was denied a loan by an automated decision-making system. We aimed to assist this individual in overcoming their difficult situation—that is, to achieve *algorithmic recourse*, which was contingent on offering answers to two questions: why and how? We studied the relation between these questions and arrived at distinct responses, namely *contrastive explanations* and *consequential recommendations*. Mindful of the goal of recourse, we emphasized *minimal* consequential recommendations over *nearest* contrastive explanations, as the former directly optimizes for the least effort from the individual. Furthermore, we noted that offering recourse recommendations automatically implied recourse explanations (through simulation of the causal effect of undertaken actions), whereas the converse would not. In reviewing the literature, however, we observed an underexploration of consequential recommendations, which we attribute to its reliance on additional assumptions at the level of the causal generative process of the world in which actions take place.

We remark that the primary emphasis of recourse is on providing actionable recommendations for overcoming adverse predictions, unlike counterfactual explanations that highlight similar instances with different predictions. This subtlety is also apparent in parallel literature on adversarial ML (see Section 3.6): counterfactual explanations identify the adversarial instances, whereas recourse informs on how to create them. Therefore, to provide actionable recommendations under real-world constraints, we must consider that actions on one subset of variables may have consequences on other variables. Such constraints can be accounted for using the SCM framework, and therefore we primarily take this approach in our work.

In addition to unifying and precisely defining recourse, we present an overview of the many constraints (e.g., actionability, plausibility, diversity, sparsity) that are needed to model realistic recourse settings. With accompanying illustrative examples, we distinguish between the notions of *dist* versus *cost*, and *plausibility* versus *actionability* (*feasibility*), whose distinctions are largely ignored in the literature. Throughout, we reiterate that these notions are individual-/context dependent, and that formulations cannot arise from a technical perspective alone. We summarize the technical literature in Table 1 as a guide for practitioners looking for methods that satisfy certain properties, and researchers who want to identify open problems and methods to further develop.

Finally, we identify several prospective research directions that challenge the assumptions of existing setups and present extensions to better situate recourse in the broader ethical ML literature. The presented examples and discussion serve to illustrate the diversity of stakeholder needs and a tension between the desirable system properties (fairness, security, privacy, robustness) that we seek to offer alongside recourse. Satisfyingly addressing these needs and navigating the entailed trade-offs may require new definitions and techniques, and relies on the cross-disciplinary expertise of a panel of technical and social scientists. It is our hope that this document may guide further discussion and progress in this direction.

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